

## Abstract

**Background:** Premenstrual dysphoric disorder (PMDD) is a DSM-5 depressive disorder affecting an estimated 31 million women and girls worldwide.<sup>1</sup> This scoping review examined psychosocial outcomes and coping mechanisms among women with PMDD and assessed the geographic distribution of the evidence.

**Methods:** Following Arksey and O'Malley's framework and PRISMA-ScR guidance, we searched PubMed/MEDLINE, CINAHL, APA PsycInfo, APA PsycArticles and the Cochrane Library for peer-reviewed studies published 2010–2025. Eligible empirical and qualitative studies had PMDD as the primary population and at least one psychosocial outcome or coping mechanism as a primary aim.

**Results:** The search identified 1,943 records; 1,618 remained after deduplication and 44 studies met the inclusion criteria. Depression was most documented (n=39), followed by psychological distress (n=24), interpersonal functioning (n=18), suicidal ideation or self-harm (n=14) and quality of life (n=14). Only three studies examined coping mechanisms. Only three studies (6.8%) originated from low- and lower-middle-income countries, including one from Nigeria.

**Conclusions:** Evidence on PMDD's psychosocial burden is growing but geographically concentrated in high-income countries, revealing a critical LMIC evidence gap.

**Keywords:** premenstrual dysphoric disorder; psychosocial outcomes; coping mechanisms; LMICs; scoping review

## Introduction, Objective, and Research Question

**Introduction:** PMDD is a recurring, hormone-linked mood disorder in which irritability, mood swings, depressed mood and anxiety emerge in the late luteal phase and remit after menses begin; DSM-5 classifies it as a depressive disorder.<sup>2</sup> Using prospectively confirmed diagnoses, community prevalence is about 1.6%, roughly 31 million women and girls worldwide.<sup>1</sup> PMDD has been linked with suicidal thoughts, strained relationships and reduced quality of life, but this evidence is scattered across disciplines, and it is unclear how women cope or where the research has been conducted.

**Objective:** To map psychosocial outcomes and coping mechanisms reported among women with PMDD and identify geographic gaps in the published literature.

**Research Question:** What psychosocial outcomes and coping mechanisms have been reported in studies of women with PMDD published between 2010 and 2025, and how are these studies distributed across geographic settings?

## Methods

| Category           | Description                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Study Design       | Scoping review guided by Arksey and O'Malley's framework <sup>3</sup> and reported using PRISMA-ScR. <sup>4</sup>                                              |
| Databases          | PubMed/MEDLINE, CINAHL, APA PsycInfo, APA PsycArticles and Cochrane Library.                                                                                   |
| Publication Years  | 2010–2025.                                                                                                                                                     |
| Inclusion Criteria | Peer-reviewed empirical or qualitative studies with PMDD as the primary population and at least one psychosocial outcome or coping mechanism as a primary aim. |
| Exclusion Criteria | Animal studies, laboratory research, reviews, interventions and non-peer-reviewed publications.                                                                |
| Screening Process  | 1,943 records identified; 325 duplicates removed; 1,618 records screened by title and abstract; 64 assessed for eligibility; 44 studies included.              |
| Evidence Mapping   | Included studies categorized by psychosocial outcome, coping mechanism and geographic setting (country and income level).                                      |

1,943

records identified  
across 5 databases

44

studies included  
in the scoping review

3

studies addressing coping  
6.8% of 44

3

low/lower-middle-income studies  
6.8% of 44

Fig 1. Study Selection — PRISMA-ScR Flow Diagram

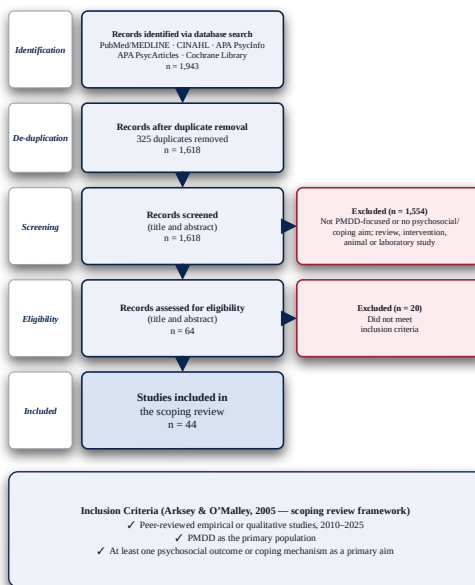
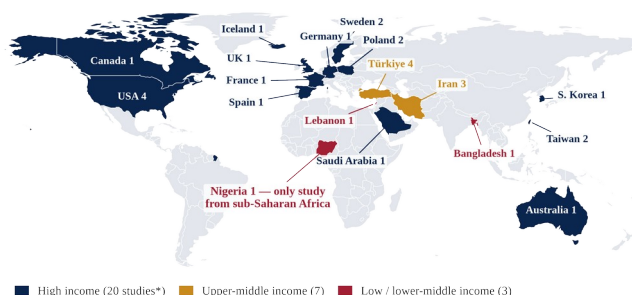


Fig 3. Geographic Setting of the 44 Included Studies

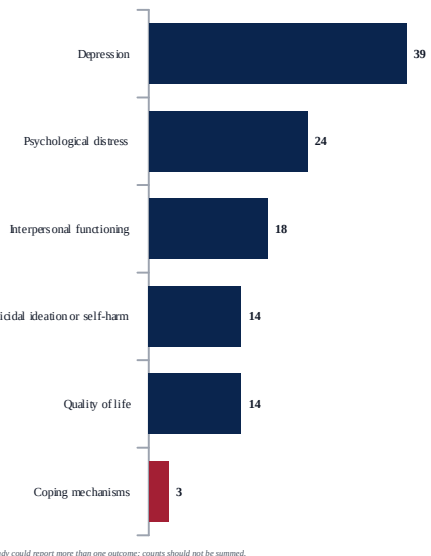


■ High income (20 studies\*) ■ Upper-middle income (7) ■ Low / lower-middle income (3)

Map shows the 30 studies with an identifiable country setting (World Bank income groups). The other 14 used online or multinational samples or had no single mappable setting in the available records.

\*Includes Hong Kong (1).

Fig 2. Included Studies Reporting Each Outcome (of 44)



A study could report more than one outcome; counts should not be summed.

## Results, Discussion, Conclusion, and Limitations

## Results

The database search identified 1,943 records. After 325 duplicates were removed, 1,618 records were screened by title and abstract; 64 were assessed for eligibility; 20 were excluded, and 44 studies were included (Fig. 1).

Depression was the most documented outcome (n=39), followed by psychological distress (n=24), interpersonal functioning (n=18), suicidal ideation or self-harm (n=14) and quality of life (n=14). Only three studies examined coping mechanisms (Fig. 2).

Geographic representation was skewed: 20 studies came from high-income countries and 7 from upper-middle-income countries (Turkey, Iran); only three (6.8%) came from low- and lower-middle-income countries (Bangladesh, Lebanon, Nigeria), and Nigeria was the only study from sub-Saharan Africa (Fig. 3).

## Discussion

PMDD research consistently documents depression, distress, relationship difficulties and suicidality; large studies have linked PMDD with suicidal ideation and attempts.<sup>5,6</sup> Yet coping mechanisms remain largely unexamined, and the scarcity of studies from low- and lower-middle-income countries and Africa leaves the burden in these settings invisible to health systems.

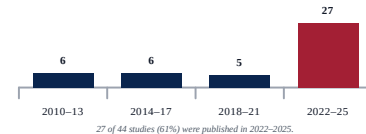
## Conclusion

Evidence on PMDD's psychosocial burden is growing but geographically concentrated in high-income countries. Findings call for culturally responsive screening, integrated mental and reproductive health services, and Africa-centred research investment.

## Limitations

- **Search scope:** studies outside the five databases or the 2010–2025 period may have been missed.
- **Study differences:** variation in design and outcome definitions limits direct comparison.
- **Coding overlap:** a study could address more than one outcome, so counts should not be summed.
- **Geographic coverage:** few studies from low- and lower-middle-income countries limit conclusions about these settings.

Fig 4. Included Studies by Publication Period



27 of 44 studies (61%) were published in 2022–2025.

## Public Health Implications and Future Work

## Public Health Implications

- **Screen for suicide risk:** ask about suicidal thoughts when women present with severe premenstrual mood symptoms.
- **Integrate care:** include PMDD screening in sexual and reproductive, school and university health services.
- **Address coping needs:** ask how women manage symptoms and what support they need.

## Future Work

- **Confirm PMDD prospectively** in studies from low- and middle-income countries.
- **Study lived experience:** stigma, coping and help-seeking in African settings.
- **Validate screening tools** in African languages.

## References



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